Stakeholder-driven municipal air-quality batch system

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Purpose and stakeholders

The proposed system is an hourly municipal air-quality data layer. Environmental analysts inspect trends, gaps and anomalous readings; city planners compare stations and policy periods; a dashboard developer requests recent values; a data steward monitors runs; and an authorised warning officer may review threshold candidates. The prototype never sends public alerts.

Evidence and limitations

The UCI Beijing dataset supplies 420,768 historical rows from 12 stations only as a scale and missing-data simulation. It cannot represent another city: calibration, precision, station siting and spatial coverage are unverified, while hourly values may be unsuitable for some health questions.

Database choice

MongoDB stores one flexible station-hour document, validated shared fields and quality flags. Compound indexes support station/time and comparison queries. PostgreSQL offers stronger fixed relational structure; files lack concurrent indexed queries; a warehouse is excessive for this operational prototype.

Plan and success criteria

The loader will use 1,000-row batches, deterministic identifiers, insert-only upserts, bounded retries, quarantine and run heartbeats. Success requires 420,768 source rows reconciled, no duplicate station-hours, a harmless replay, visible quality counts, and four stakeholder queries within a 2-second local target. Any silent imputation, unreconciled row, failed run, unapproved alert or missed target is failure.

Sources: UCI Machine Learning Repository, Beijing Multi-Site Air Quality, https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data; MongoDB, Data Modeling, https://www.mongodb.com/docs/manual/data-modeling/